Politecnico di Milano

Orbital Mechanics

Academic Year 2025/2026

Project group n.: 2553

Team members:

	Surname	**Name**	
	Frescura	Simone	
	Pifano	Nicola	
	Puzzi	Luca	
	Veronese	Lorenzo	

Contents

1 Interplanetary explorer mission **3**
1.1 Introduction . . . 3
1.2 Additional constraints . . . 3
1.3 Time window . . . 4
1.4 Initial condition . . . 5
1.5 Finding the best solution . . . 5
1.6 Solution analysis . . . 6

2 Planetary explorer mission **7**
2.1 Nominal Orbit . . . 7
2.1.1 Orbital Parameters . . . 7
2.2 Ground Track Analysis . . . 8
2.2.1 Unperturbed Ground Track . . . 8
2.2.2 Repeating Ground Track . . . 8
2.3 Orbit Propagation with Perturbations ($J_2 + \mathrm{SRP}$) . . . 9
2.3.1 Perturbed Ground Track ($J_2 + \mathrm{SRP}$) . . . 10
2.4 Orbital Elements History and Method Comparison . . . 11
2.4.1 Evolution Analysis ($J_2 + \mathrm{SRP}$) . . . 11
2.4.2 Method Comparison . . . 11
2.5 3D Visualization . . . 12
2.6 Filtering of High Frequencies . . . 13
2.7 Real Data Analysis . . . 13
2.7.1 Object Selection . . . 13
2.7.2 Data Processing . . . 14
2.7.3 Analysis of Evolution and Comparison with Theory . . . 14
2.7.4 Conclusions . . . 14

1 Interplanetary explorer mission

1.1 Introduction

The objective of assignment 1 is to perform the preliminary design of an interplanetary mission, going from a planet to an asteroid and using a fly-by to help the spacecraft reach the speed needed to cover the established distance. In our case, these are the conditions given by the scientific team:

- Departure planet: Venus
- Fly-by planet: Mars
- Arrival asteroid id: 184430
- Earliest departure: 00:00:00 01/01/2030
- Latest arrival: 00:00:00 01/01/2060

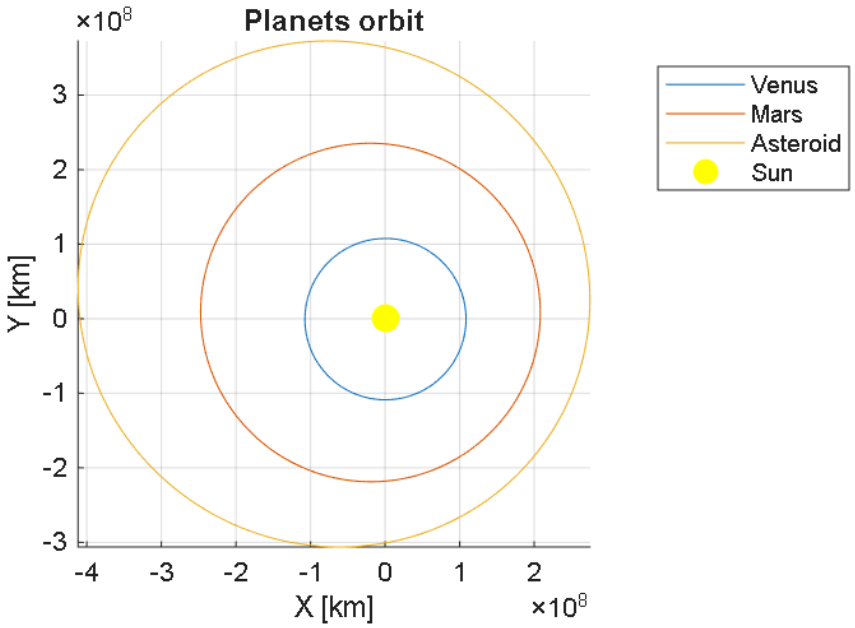


Figure 1: Orbits of the planets in the Sun-centred ecliptic frame, seen from above.

1.2 Additional constraints

First of all, in addition to the time constraint written above, we must also take into account a constraint on the altitude of the fly-by. In order to have a feasible manoeuvre, the radius of pericentre of the incoming and outgoing hyperbolas has to be greater than the radius of Mars. Moreover, the spacecraft should dodge the planet's atmosphere, in order to avoid being slowed down by drag forces and prevent related heating problems. For this reason, we set the minimum altitude of fly-by to 230 km[1], meaning the minimum radius of pericentre is set to 3789.9 km.

[1]According to [1], the exobase, that is the interface between the thermosphere and the exosphere, reaches a maximum altitude of 230 km, at which the atmosphere becomes collisionless.

1.3 Time window

The first thing our team worked on is the restriction of the time window. In fact, doing a detailed analysis for a time span of thirty years would be computationally very expensive, which is why this was chosen as the first step of the design process. First, we analysed the orbital period of the three celestial objects: $T_V = 0.616$ years, $T_M = 1.882$ years, $T_A = 3.528$ years. However, sidereal periods do not provide enough information to really understand the problem. What is needed is the synodic period, which can be expressed as follows [2]

$$T_{syn} = \frac{T_1 T_2}{|T_1 - T_2|}. \tag{1}$$

As the equation states, the synodic period is only defined for pairs of orbiting objects. In addition, this formula is valid exclusively for circular and coplanar orbits: since ours have only a few degrees of inclination difference and the eccentricities aren't high, we think the assumption is allowed at this preliminary stage. These are the values obtained in our case:

- $T_{syn} = 0.915$ years between Mars and Venus
- $T_{syn} = 0.746$ years between the asteroid and Venus
- $T_{syn} = 4.034$ years between the asteroid and Mars.

Identifying an equivalent synodic period of the configuration of the three celestial objects would allow us to determine a smaller time window than the starting one, knowing there would be no loss of information, since every possible configuration would take place in that reduced time span and would then repeat. However, at this stage we have no evidence that an equivalent synodic period of the three bodies exists and lasts less than thirty years. Moreover, the process would require a certain degree of approximation, of which we don't know the entity. For these reasons, we decided to follow a different approach.

The expression for the total velocity cost of the mission is

$$\Delta v_{tot} = \Delta V_1 + \Delta V_2 + \Delta v_p \tag{2}$$

where ΔV_1 is the impulse needed to go from Venus' orbit to the first interplanetary arc, ΔV_2 is the impulse needed to go from the second interplanetary arc to the asteroid's orbit, and Δv_p is the cost of the manoeuvre at the pericentre of the fly-by. Now, the idea is to scan the full time window with a coarse grid search in order to define a window where the total delta velocity is reasonable. To do this, we implemented 3 nested *for* loops with twenty points each, meaning we have a resolution of 1.5 years, and we computed the total delta velocity at each point. Here, the big design choice is the threshold value above which we discard the Δv. By analysing the cost of the two Lambert arcs via the porkchop plots and assuming a reasonable Δv_p for the powered manoeuvre at pericentre, after taking some margin we defined a threshold value of 25 km/s. We can now select the right departure, fly-by and arrival windows by taking the earliest and latest time at which the velocity constraint is respected. In this way we get these convenient time windows:

- departure between 11:22:06 on 19/8/2042 and 05:03:09 on 12/12/2048
- fly-by between 16:25:15 on 18/3/2044 and 20:12:37 on 08/02/2052

[2] Formula taken from [2]

- arrival between 18:56:50 on 16/10/2045 and 16:25:16 on 07/09/2053

1.4 Initial condition

Now that we know a good time window, our task is to find the optimal time of departure, fly-by and arrival in order to minimize the cost of the mission. The tool we used is the MatLab® function *fmincon* [3], which takes as input the function to minimize, the boundaries that the variables have to respect and an initial guess for the decision variables. The function is designed to find local minima of the input function, which means that it is not meant to work globally. This has an impact on our work, because it gives importance to the first guess we feed to the function. In fact, when we first tried some random first guesses, we always got different solutions.

Departure	**Fly-by**	**Arrival**	Δv_{tot}
13/05/2043	14/05/2044	23/08/2046	31.4 km/s
29/09/2045	12/02/2048	29/09/2049	15.5 km/s
16/11/2048	07/02/2052	07/09/2053	17.8 km/s

We solved the problem by doing another grid search within the selected time window, looking for the initial condition corresponding to the best delta velocity we can find. This time we decided to go for a finer grid, using 30 points per loop. The result we get is

- departure at 16:46:10 on 02/05/2046
- fly-by at 13:17:06 on 09/01/2048
- arrival at 02:55:06 on 01/05/2049.

1.5 Finding the best solution

Now that we know the time boundaries and the best initial guess, we just need to run *fmincon* to get the times at which the mission is the cheapest. But let's take a step back and first explain how we set up the function that computes Δv_{tot}:

1. Use of given functions *uplanet* and *ephAsteroid* to have the ephemerides of the celestial objects and retrieve their orbital speeds
2. Use of given function *lambertMR* to compute the heliocentric velocities of the two arcs
3. Compute the hyperbolic velocities v_{inf}^- and v_{inf}^+
4. Calculate the turn angle δ
5. Set the constraint on the radius of pericentre by computing δ_{max} from r_p^{min}
6. Find the radius of pericentre r_p through the function *fzero*
7. Compute the delta velocity at pericentre Δv_p
8. Compute Δv_{tot} as in equation (2).

Through this procedure we obtained the following solution:

$$\Delta v_{tot}^{min} = 13.026\,\text{km/s} \tag{3}$$

with

- departure at 01:30:51 on 27/05/2046
- fly-by at 08:00:14 on 18/03/2048
- arrival at 17:21:48 on 04/09/2049.

1.6 Solution analysis

Let's further analyse that Δv_{tot}^{min} by highlighting the different phases of the mission:

$$\Delta V_1 = 8.919\,\text{km/s} \tag{4}$$

$$\Delta V_2 = 4.107\,\text{km/s} \tag{5}$$

$$\Delta v_p = 7.897 \cdot 10^{-5}\,\text{km/s} \tag{6}$$

From (6) we can see that the fly-by is almost unpowered, being the delta velocity of the manoeuvre very low. Looking at the velocity change due to the fly-by,

$$\Delta v_{fb} = 1.013\,\text{km/s} \tag{7}$$

we notice that it is bigger than the impulse at pericentre, but still lower than the other two, meaning that the gravity assist has a minor, but still relevant, effect on the mission.

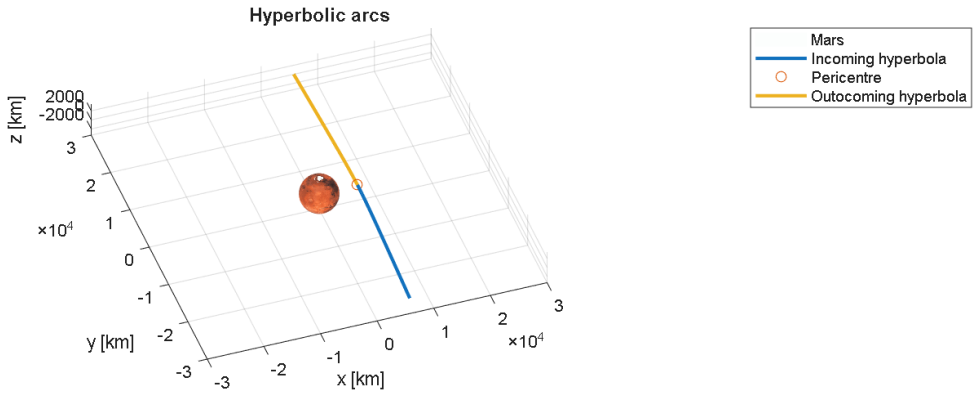


Figure 2: Trajectory of the fly-by in the Mars centred inertial reference frame

Δv_{fb} is low because the satellite passes quite far from the planet, as we can see from the radius of the pericentre

$$r_p = 6764.299\,\text{km}, \tag{8}$$

which means that the passage takes place at an altitude of 3374.399 km. This is the reason why also the turn angle is low

$$\delta = 4.85\,^\circ. \tag{9}$$

Far from the planet speeds are lower too, affecting the duration of the fly-by. Considering a finite sphere of influence with radius of 570000[3] km, the fly-by lasts approximately 1 day, 2 hours and 23 minutes.

[3]We took the ratio between the radius of the sphere of influence and the mean orbital radius of Mars from the slides of lab chapter 4. The value is $\frac{r_{SOI}}{r_M} = 0.0025$.

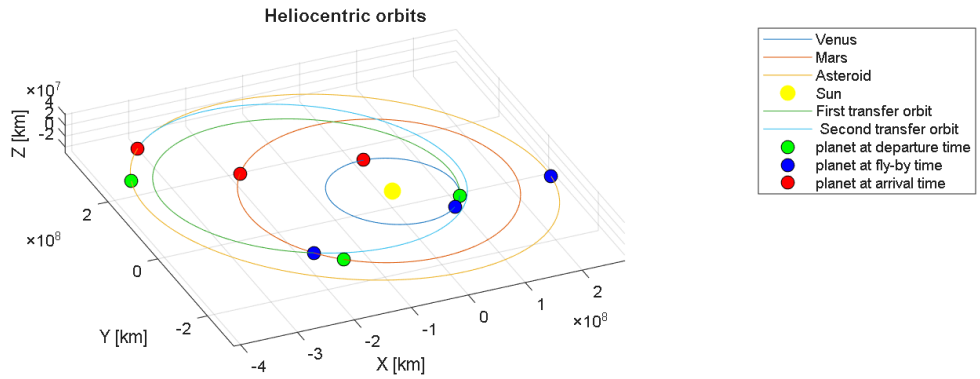


Figure 3: Heliocentric trajectory of the spacecraft in the Sun-centred ecliptic reference frame. Orbits and positions of planets at characteristic times are also depicted.

Let's now look at the heliocentric trajectory depicted in figure 3. From the start of the mission, the asteroid makes almost a full revolution around the Sun. In fact, the time of flight of the mission is

$$tof = 3.28\,\text{years}, \tag{10}$$

nearly the length of an orbital period of the asteroid.

These are the orbital elements that characterize the two transfer arcs.

Arc	**a**	**e**	**i**	Ω	ω
Venus to Mars	2.48e+08 km	0.5602	3.58°	42.33°	292.72°
Mars to Asteroid	2.71e+08 km	0.5563	3.53°	42.44°	284.48°

It is easy to notice that the two ellipses are very similar to each other, due to the fact that the fly-by does not change the speed significantly. The angle covered by the two arcs is also comparable: the first transfer angle is equal to 239.89°, while the second is 289.20°.

In conclusion, we found a feasible and efficient solution that satisfies all constraints and allows us to complete the mission.

2 Planetary explorer mission

2.1 Nominal Orbit

2.1.1 Orbital Parameters

The analysis focuses on the orbit of a satellite around Venus, defined by the following Keplerian elements in the planetocentric equatorial inertial frame:

Table 1: Assigned Orbital parameters

t_0	a [km]	e [-]	i [°]	Ω [°]	ω [°]	θ_0 [°]
2045/08/25	341475	0.0142	149	46	94	0

The orbit is retrograde ($i > 90°$) and quasi-circular ($e \approx 0.01$). The resulting orbital period is $T \approx 25.46$ days, with a perigee altitude $h_p \approx 330,574$ km.

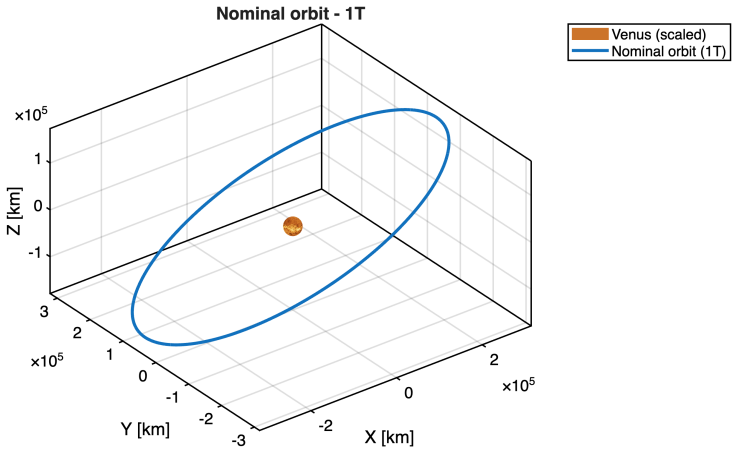


Figure 4: Nominal Orbit

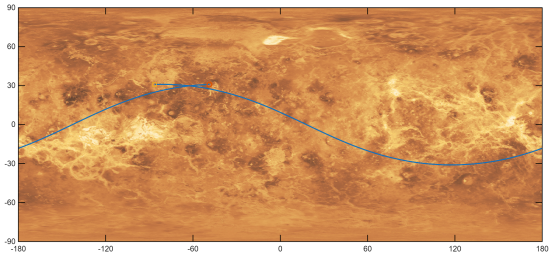
2.2 Ground Track Analysis

2.2.1 Unperturbed Ground Track

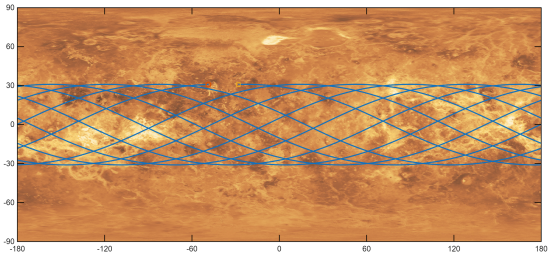
The ground track was generated by propagating the Keplerian motion and transforming the inertial position into the planet-fixed frame. Venus's retrograde rotation has been taken into account. The angular velocity of the planet was modeled as $\omega_{GT} = -\omega_{planet} > 0$ to ensure correct longitudinal displacement.
The ground tracks were plotted for 1 and 9 periods: in fact, after 9 periods, the ground track has a small shift with respect to the first ground track. This means that the spacecraft has covered the whole area ranging from the higher to the lower latitude and from the eastern to the western longitude with a good resolution. Considering a real mission scenario, in this case the satellite's camera would have been able to scan the entire surface of the planet between the maximum and minimum latitude.
The generated ground tracks were validated against theoretical expectations: the track is confined between $\varphi \approx \pm 31°$, consistently with the retrograde inclination constraint $\varphi_{max} \approx 180° - i = 31°$.
After one period T, the ground track shifts eastward by $\Delta\lambda \approx 38°$. This matches the theoretical prediction based on the ratio between the orbital period and Venus's rotation period, confirming the correct implementation of the retrograde rotation sign.

2.2.2 Repeating Ground Track

The repeating ground track was designed by modifying the semimajor axis a to satisfy the condition $kT_{orb} = m|T_{rot}|$.



(a) Nominal GT over 1 period



(b) Nominal GT over 9 periods

Figure 5: Unperturbed Ground Tracks

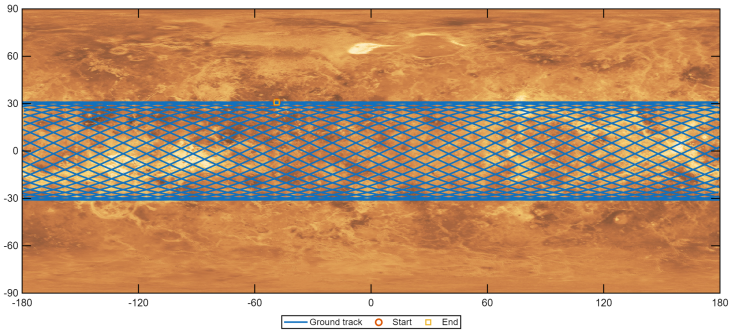


Figure 6: Repeating Ground Track

Given the ratio $\frac{k}{m} = \frac{30}{1}$, the new semimajor axis obtained is $a_{rep} \approx 159150$ km, with a repetition cycle of $T_{cycle} \approx 243$ days leading to a perfect closure in the 2BP model:

$$\Delta\lambda \approx 0^\circ, \quad \Delta\varphi \approx 0^\circ$$

2.3 Orbit Propagation with Perturbations (J_2 + SRP)

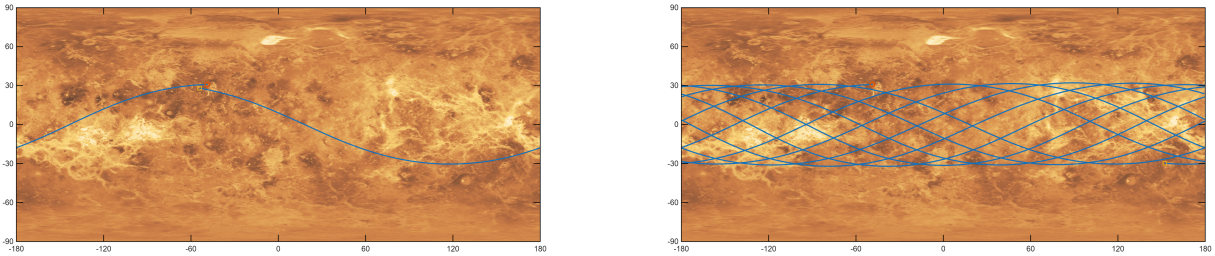
The orbit was then propagated taking into account the two perturbations assigned: J_2 and Solar Radiation Pressure.

The J_2 perturbation is a consequence the planetary oblateness and it is expected to induce secular drifts primarily on Ω and ω.
The SRP perturbation has been modelled assuming the shape of the spacecraft to be an equivalent sphere, in order to have a constant surface exposed to the solar radiation (Cannonball model[2]). In this way the force applied is oriented along the direction Sun-spacecraft. A coefficient of reflectivity of $C_r = 1.0$ and an Area over Mass ratio of $\frac{A}{M} = 5.4766$ m^2/kg were assigned to properly model the SRP. The acceleration induced by this perturbation is proportional to:

$$\mathbf{a}_{SRP} \propto -C_r \frac{A}{M} \hat{\mathbf{s}}. \tag{11}$$

The Sun-Venus vector $\hat{\mathbf{s}}$ is defined accounting for Venus' axial tilt. Eclipse effects were neglected because the spacecraft operates in a high-altitude orbit, where the duration of shadow events is negligible compared to the orbital period.

Two consistent numerical approaches were implemented:



(a) Perturbed GT over 1 period (b) Perturbed GT over 9 periods

Figure 7: Perturbed Ground Tracks

- the Cartesian Method, which consists in a direct integration of the state $\dot{\mathbf{y}} = [\mathbf{v}; \mathbf{a}_{2BP} + \mathbf{a}_{J_2} + \mathbf{a}_{SRP}]$
- the Gauss Method, which consists in the integration of Gauss's planetary equations for the state $[a, e, i, \Omega, \omega, M]$.

The perturbing accelerations are projected into the RSW(radial – transversal – out-of-plane) local frame. A safety threshold $e_{min} = 10^{-6}$ was implemented to avoid singularities in the equations (though never triggered as $e > e_{min}$ for 100% of the simulation). Conversions $\theta_0 \to M_0$ and $M(t) \to \theta(t)$ were performed via Kepler's Equation.

Both methods use a common time grid to avoid interpolation errors during comparison:

- Duration: 100 nominal orbits ($T_{end} \approx 255$ days)
- Solver: `ode113` with relative and absolute tolerances set to 10^{-12}.

2.3.1 Perturbed Ground Track (J_2 + SRP)

The ground track analysis was repeated including the 2 perturbations. Looking at the plots in figure 5 figure 7, it is evident in both the nominal and perturbed cases that, as the number of orbital periods increases, the differences between the various ground tracks become increasingly pronounced. This drift is associated with the effect induced by J_2 secular precession of the orbital plane (Ω) and apsidal line (ω), while the SRP alters the eccentricity (e) and the semi major axis (a) influencing the mean motion of the satellite.

Those same effects can be observed if we try to apply the same repeating ground track solution to the perturbed case. As we can see from figure 9, the ground track does not close. After the theoretical repetition cycle ($k = 30$ orbits), the total drift is:

$$\Delta\lambda \approx +105.3^\circ, \quad \Delta\varphi \approx -35.0^\circ$$

This result is given by a systematic drift ($\approx 3.5^\circ$ per orbit) that contributes to desynchronize the resonance between the spacecraft and the planet. In order to restore the repetitiveness it would be necessary to re-design the orbit accounting also for the perturbation effects or to adopt station keeping and maintenance solutions.

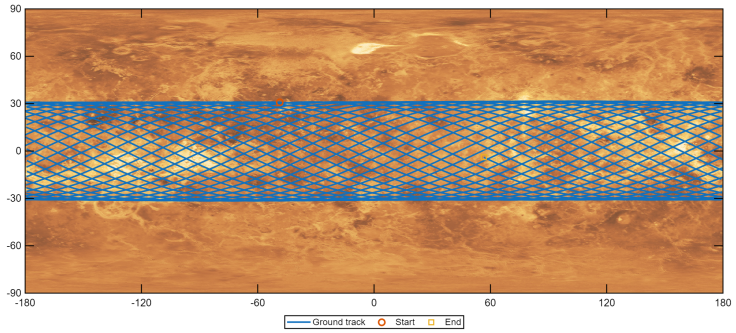


Figure 8: Repeating Perturbed Ground Track

2.4 Orbital Elements History and Method Comparison

In order to ensure physical readability, the time axis is expressed in days and angular elements are plotted in degrees.

2.4.1 Evolution Analysis ($J_2 + \mathbf{SRP}$)

The results from Cartesian and Gaussian Methods are visually superimposable. The observed evolution is consistent with theoretical predictions:

- **Semimajor Axis** $a(t)$**:** shows distinct periodic oscillations ($\Delta a \approx 3 \cdot 10^4$ km) but no secular drift. This confirms that J_2 is conservative, while SRP acts as a non-conservative external force exchanging orbital energy over the solar cycle.
- **Eccentricity** $e(t)$**:** exhibits a periodic variation over approximately a Venusian year, but another periodic trend is visible with a period around 7500 days. This high sensitivity is expected due to the very high Area-to-Mass ratio and the weak gravitational pull at high altitudes.
- **Inclination** $i(t)$**:** remains stable with small oscillations ($\Delta i \approx 2°$) on the short and long period, confirming that out-of-plane perturbations are limited, but the difference increases going from $\approx 149°$ to $\approx 156°$ over 500.
- **RAAN** $\Omega(t)$**:** affected by small oscillation on the short period, but displays a strong positive secular drift.
- **Arg. of Perigee** $\omega(t)$**:** shows secular precession. A rapid "jump" is observed at $t \approx 220$ days (around a Venusian year); this corresponds to the instant where eccentricity reaches its minimum ($e \to 0$), causing the pericenter location to become geometrically undefined (a mathematical singularity, not a physical discontinuity).

2.4.2 Method Comparison

The results of the 2 methods are comparable as their relative errors are negligible on the short periods and grows over time due to the natural accumulation of small numerical variations. Specifically the semimajor axis error goes from $\sim 10^{-11}$ for a propagation of 10 periods to 10^{-3} for 500 periods, the eccentricity error from $\sim 10^{-11}$ to 10^{-9}. The inclination and the right ascension of the ascending node go from $\sim 10^{-10}$ deg to 10^{-7}. The argument of the perigee error stays around $\sim 10^{-10}$ deg and jump up to values of $\sim 10^{-5}$ deg only at the $e \to 0$ singularity for both propagation time.

From the computational cost point of view, as expected the Cartesian method is slightly faster with respect to the Gauss method, due to fact that the latter require more complex calculations and coordinate transformations at each integration step. The computational cost varies at each run so the values reported here corresponds to the mean value of 10 consecutive runs after a warm up phase:

- Cartesian : ~ 3.087 s.
- Gauss: ~ 3.362 s.

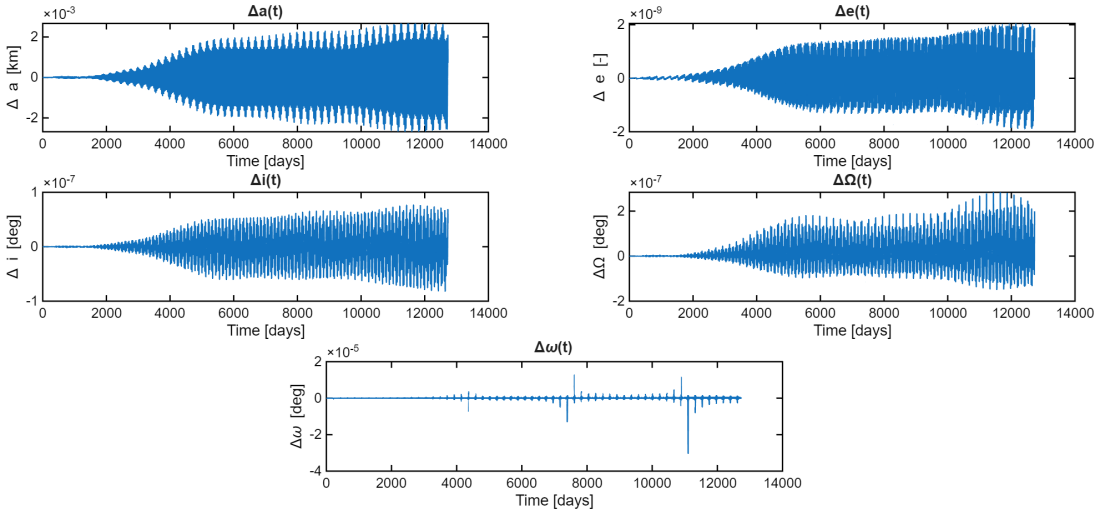


Figure 9: Relative Error

2.5 3D Visualization

In 10 represents the 3D trajectory plot over 100 orbits(for visual clarity), where a time-based colour map is employed to visualize the orbital evolution. The colours of the orbits start with cool tones and progressively shift to warm tones over time.

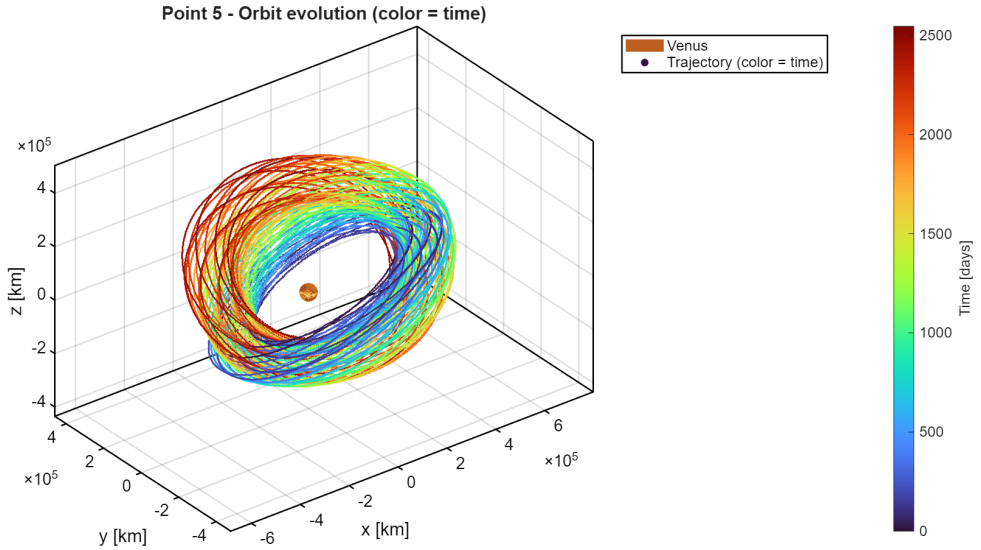


Figure 10: 3D Orbit Evolution

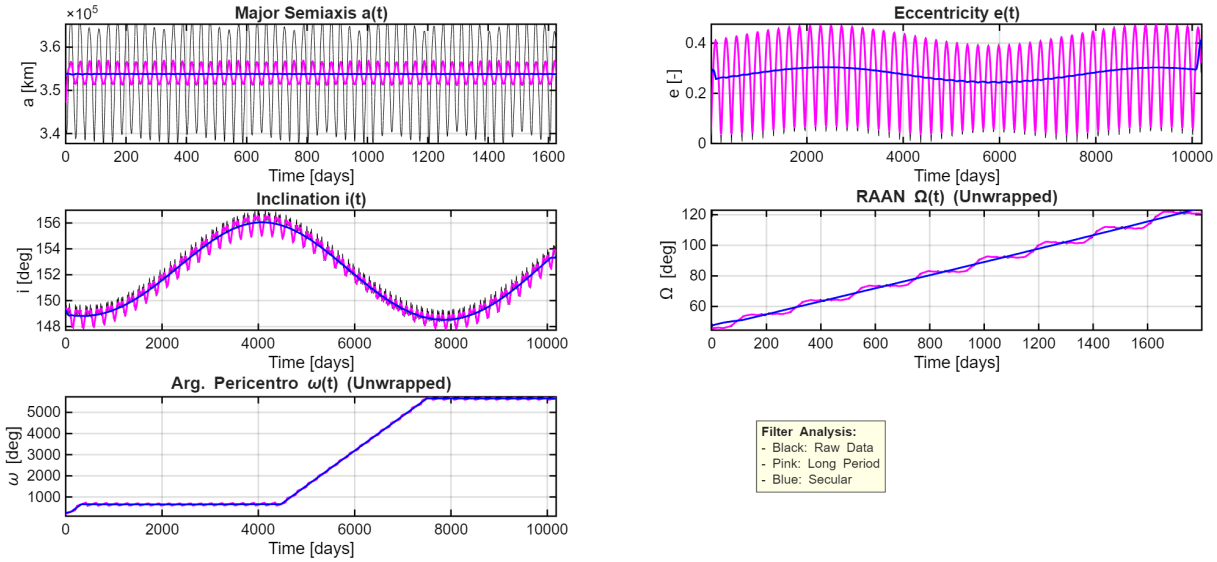


Figure 11: Filtered Parameters

2.6 Filtering of High Frequencies

As we can observe from the plots in figure 11, the raw output from the propagation contains high-frequency oscillations mainly induced by the J_2 zonal harmonic, which repeat with the frequency of the satellite's orbital motion.
In order to focus on the long-term behaviour of the orbit and find secular trends, a post-processing low-pass filter was applied to the history of the Keplerian elements obtained. The data filtering was performed using a moving average technique applied with two distinct window sizes. The first window corresponds to the satellite's orbital period (T_{orb}), selected to smooth out short-period oscillations. A second filter, with a window size corresponding to one Venusian year (≈ 224.6 days), was implemented to isolate and highlight the secular trends of the orbital elements. The analysis of the filtered data shows distinct secular behaviors for each orbital element. The semi-major axis maintains a constant secular mean, consistent with the conservative nature of the J_2 perturbation and the cyclic energy exchange characteristic of SRP, which averages to zero over long periods. In contrast, the eccentricity is dominated by a yearly oscillation driven by the solar geometry; filtering this frequency reveals an underlying ultra-long period trend of approximately 7500 days, which is visible also in the inclination, in the right ascension of the ascending node, and in the argument of perigee, suggesting a dynamical coupling between these parameters. The right ascension of the ascending node also displays a monotonic growth typical of J_2-induced nodal precession in retrograde orbits. Finally, the argument of perigee exhibits a peculiar "staircase" secular evolution, where rapid variations occur whenever the eccentricity approaches zero, superimposed on the expected apsidal drift.

2.7 Real Data Analysis

2.7.1 Object Selection

To analyze the real-world evolution of orbital elements, two uncontrolled objects were selected from the USSPACECOM catalog [4](*Space-Track*). The objects reside in distinct orbital regimes to highlight different dominant perturbations:

- NORAD 13819 (SL-3 R/B): a rocket body in LEO orbit, chosen to highlight the effects of atmospheric drag.
- NORAD 8820 (LAGEOS-1): a spherical satellite in MEO orbit, chosen to focus on gravitational perturbations.

2.7.2 Data Processing

TLEs were downloaded for a three-month interval (22 September – 20 December 2025). The raw data was processed to extract the time history of the semimajor axis (a), eccentricity (e), inclination (i), Right Ascension of the Ascending Node (Ω), and Argument of Perigee (ω). The angular elements were unwrapped to allow for a linear regression analysis of their secular trends.

2.7.3 Analysis of Evolution and Comparison with Theory

LEO regime analysis (NORAD 13819): the plotted data reveals a clear decay in the semimajor axis ($\Delta a \approx -9$ km over 3 months). This behaviour perfectly aligns with the theoretical prediction for the atmospheric drag perturbation. In LEO, the atmosphere dissipates orbital energy, causing the orbit to shrink and circularize.
The angles Ω and ω exhibit a linear drift. This aligns with the J_2 perturbation theory (Earth's oblateness), which predicts constant precession rates for a given semimajor axis and inclination.

MEO Regime Analysis (NORAD 8820): the plotted data shows that semimajor axis remains constant. This aligns with the theory that atmospheric density in MEO is effectively zero; thus, the drag force vanishes, and orbital energy is conserved. The evolution is dominated purely by the linear precession of Ω and ω. The observed rate of nodal precession ($\dot{\Omega} > 0$) is consistent with the theoretical J_2 effect for a retrograde orbit ($i > 90°$).

2.7.4 Conclusions

Overall, the real data aligns strongly with the theoretical models of drag and J_2: for the LEO object (13819), measurable orbital decay (drag) and marked secular precessions (J_2) are evident; for the MEO object (8820, LAGEOS-1), an extremely stable orbit in terms of semi-major axis is observed, while Ω and ω evolve with secular rates compatible with J_2. However, small deviations and oscillations are visible in the eccentricity and inclination plots. These can be attributed to: the nature of TLEs which provide "mean" elements that smooth out short-period oscillations but contain inherent reconstruction errors and noise compared to precise GPS ephemerides but also unmodelled perturbations such as Luni-Solar perturbations and higher-order Earth harmonics.

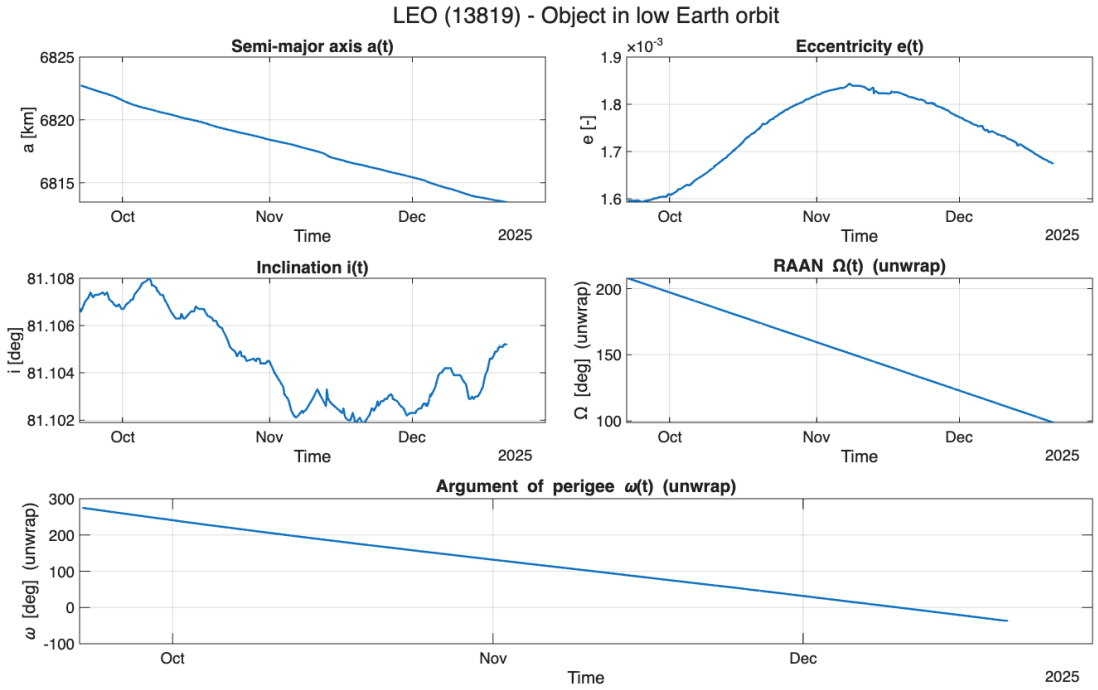


Figure 12: NORAD 13819 (SL-3 R/B).

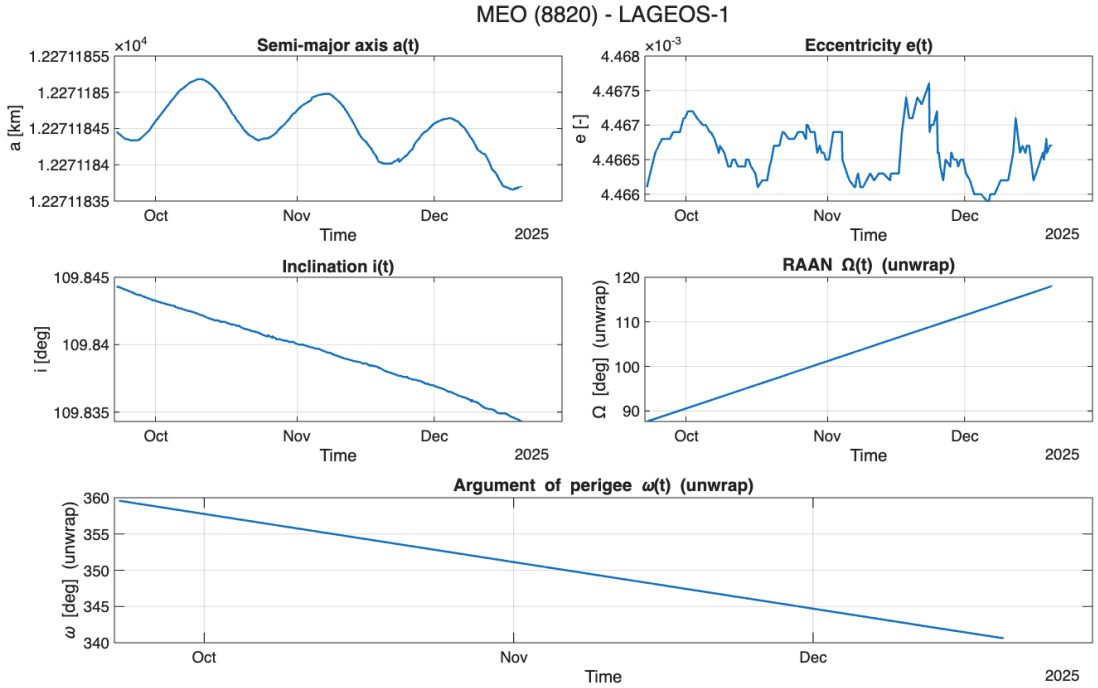


Figure 13: NORAD 8820 (LAGEOS-1)

References

[1] S. W. Bougher, T. E. Cravens, J. Grebowsky, and J. Luhmann, "The aeronomy of mars: Characterization by maven of the upper atmosphere reservoir that regulates volatile escape," *Space Science Reviews*, vol. 195, pp. 423–450, 2014. DOI: 10.1007/s11214-014-0053-7. [Online]. Available: https://lasp.colorado.edu/maven/files/2014/10/MAVEN_Aeronomy-of-Mars.pdf.

[2] H. D. Curtis, *Orbital Mechanics for Engineering Students*, 3rd ed. Oxford: Elsevier Butterworth–Heinemann, 2014.

[3] MathWorks, Inc. "Fmincon – find minimum of constrained nonlinear multivariable function." Accessed: 2025-01-05, MathWorks. [Online]. Available: https://it.mathworks.com/help/optim/ug/fmincon.html.

[4] United States Space Command, *Space-Track.org Satellite Catalog*, https://www.space-track.org/#catalog, Accessed: 2026-01-06, 2026.